

TPC-399: Cross-Family Replication of a Finite C1 Endpoint Microgrid

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Abstract

TPC-398 reported a finite endpoint microgrid on one affine family: the interior probes through $\lambda = 31/32$ were origin-stable, whereas the endpoint $\lambda = 1$ was not. TPC-399 tests the smallest natural continuation, namely a second coordinate-disjoint family with the same response-blind protocol. Its 96-row panel compares each current same-law cohort mean with a hash-locked all-origin mean from TPC-398. Every one of the 16 law/normalization cells passes both the three-percent calibration and holdout transfer caps, and all 16 within-family transfers pass. At the same time, the endpoint $\lambda = 1$ fails the one-percent origin-spread rule in all four normalizations, with maximum spreads between 0.0622195 and 0.0625497. The result is a finite cross-family replication together with a finite obstruction to inferring origin uniformity from mean transfer. It is not an arithmetic, asymptotic, Route-A/Route-B, or twin-prime theorem.

1 Question and claim boundary

The preceding TPC-398 audit used a response-blind grid of four finite matrix probes and found a diagnostic split: origin stability persisted through $31/32$, but the endpoint lost stability. The present paper asks whether this split is reproducible on a fresh family, while keeping the parent comparison frozen before reading the new response.

All objects below are finite proxy objects. In particular, a fractional coefficient in a matrix interpolation is not asserted to be an arithmetic character, a prime sign law, or a source weight. The claim firewall is

$$\text{ARITHMETIC_ADVANCE=NO,} \quad \text{FIXED_POWER_CREDIT=0,} \quad \text{FULL_GATE_B=OPEN.}$$

The official Session evaluator files are absent from this checkout, so the local proof, independent checker, and Bridge-B artifacts provide finite consistency evidence only.

2 Finite construction

For a prime $p \in (Q, 2Q]$, set $Q = 8192$, $H = 66$, exponent one, and $\beta = 2$. The component used in the certificate is

$$K_p(u, v) = p(p/Q)^2 \frac{H^2}{H^2 + (u - v)^2} \left(\mathbf{1}_{p|u-v} - \frac{1}{p-1} \right) \\ \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

Let

$$G(u) = \sum_{p \in (Q, 2Q]} \sum_v K_p(u, v)^2.$$

If the prime shell is indexed increasingly, define

$$M_+ = \sum_p K_p, \quad M_{\text{alt}} = \sum_p (-1)^{\text{index}(p)} K_p.$$

The four probes are the exact finite combinations

$$M_\lambda = (1 - \lambda)M_+ + \lambda M_{\text{alt}}, \text{quad } \lambda \in \left\{ \frac{7}{8}, \frac{15}{16}, \frac{31}{32}, 1 \right\}.$$

Table 1: TPC-399 finite panel. Counts are across the four laws.

normalization	origin	cross-cal	cross-hold	within	max R
local diagonal	3/4	4/4	4/4	4/4	0.0622194516
pooled train scalar	3/4	4/4	4/4	4/4	0.0625496889
origin scalar	3/4	4/4	4/4	4/4	0.0625446188
frozen train-1024 scalar	3/4	4/4	4/4	4/4	0.0625496889

This identity is linear algebra and is checked separately with exact rational arithmetic at a small anchor; it is not a claim about an underlying arithmetic interpolation.

The fresh candidate family is

$$a_j = 7200001 + 401j, \quad 0 \leq j < 41,$$

with selected indices (0, 8, 16, 24, 32, 40). The origins are

$$7200001, 7203209, 7206417, 7209625, 7212833, 7216041.$$

The first three are calibration origins and the last three are holdout origins. Each window has $N = 1024$ points, split into eight blocks of length 128. The fixed $c = 3$ band retains block pairs whose block-index difference is at most three.

3 Diagnostics and frozen parent interface

We record four normalizations: local diagonal, pooled calibration scalar, current-origin scalar, and first-calibration frozen scalar. If $S_\lambda(o)$ denotes the resulting masked spectral diagnostic, its declared origin spread is

$$R_\lambda = \frac{\max_o S_\lambda(o) - \min_o S_\lambda(o)}{\text{mean}_o S_\lambda(o)}.$$

The origin cap is $R_\lambda \leq 0.01$. Finite spectral and Schur caps are 0.64 and 0.83, respectively.

TPC-398 is imported only through its hash-locked code and canonical certificate. For each law and normalization, let $B_\lambda^{(398)}$ be its recorded all-origin mean. The current cohort errors are defined directly by

$$E_{\text{cal}} = \frac{\text{mean}_{o \in \mathcal{C}} S_\lambda(o)}{B_\lambda^{(398)}} - 1, \quad E_{\text{hold}} = \frac{\text{mean}_{o \in \mathcal{H}} S_\lambda(o)}{B_\lambda^{(398)}} - 1.$$

The cap for both is 0.03. No segment interpolation or response-dependent fit is performed. The within-family transfer error is

$$E_{\text{within}} = \frac{\text{mean}_{o \in \mathcal{H}} S_\lambda(o)}{\text{mean}_{o \in \mathcal{C}} S_\lambda(o)} - 1,$$

also tested at cap 0.03.

The parent locks are

```
code: 10c55a6b9e3c4dc11674780f7d1d98508d223729e18ffde7f27a88d7790a3382
certificate: 3f944db8218d8c18a2f2c756dcdf26483afe18e4ac681695beba56b222256150
```

The producer accumulates the shell in ascending order. An independent checker accumulates it in descending order, reconstructs the matrices, and recomputes all aggregates without importing producer functions. At the exact anchor [7200001, 7200014) with $Q = 8$ and shell {11, 13}, ‘Fraction’ arithmetic verifies positive geometry, symmetry, and all four interpolation identities.

4 Results

The interior laws 7/8, 15/16, and 31/32 pass the one-percent origin-spread cap under all four normalizations. The endpoint law 1 passes none, giving 12 of 16 origin-stable cells. The maximum absolute cross-family calibration errors, in local, pooled, origin, and frozen order, are

$$0.0109155432, \quad 0.0105206423, \quad 0.0105131815, \quad 0.0105299678,$$

and the corresponding holdout maxima are

$$0.0027174217, \quad 0.0021628845, \quad 0.0021636109, \quad 0.0021721329.$$

All 16 cross-family calibration cells and all 16 cross-family holdout cells pass. The maximum absolute within-family transfer errors in the same order are 0.0081096008, 0.0082707443, 0.0082627033, and 0.0082707443; all 16 pass. No spectral or Schur row fails among the 96 rows.

5 Interpretation and route ledger

The strongest positive result is a response-blind second-family replication: the 16 frozen same-law TPC-398 means transfer to the new family at the declared three-percent scale in both cohort roles. The strongest obstruction is that this transfer does not remove endpoint origin instability: at $\lambda = 1$, the maximum spread remains 6.22–6.25 percent. Thus mean transfer and origin uniformity are empirically separate finite diagnostics.

The reusable structure is a direct hash-locked same-law interface, exact finite matrix interpolation, coordinate-disjoint calibration/holdout families, reverse-order replay, and separate origin and transfer gates. The next response-blind clue is

ROUND2_CLUE=TEST_C1_ENDPOINT_MICROGRID_THIRD_FAMILY_REPLICATION.

The theorem ledger is intentionally conservative:

status	finite scope
PROVED_EXACT_Finite	selection, disjointness, hashes, anchor identities
NUMERICALLY_CERTIFIED	96 rows and stated aggregate flags
OPEN	source-valid uniformity, growing bounds, arithmetic L^2
NONE	arithmetic advance, fixed-power credit, twin-prime result

Route-A is not officially evaluated because its Session evaluator is absent; Route-B remains open. The certificate makes no claim about a source-uniform estimate, an arithmetic L^2 bound, a growing operator, or the twin-prime conjecture.

Reproduction

The project directory contains the required README, code, experiments, results, notes, proof package, and Bridge-B artifacts. The canonical certificate is ‘results/tpc399_certificate.json’. The release contains byte-identical ‘paper/main.pdf’ and ‘paper/paper.pdf’. The README lists normal and optimized commands for the producer, independent checker, mutation stress test, and Bridge-B checker.